

Ledjo Lleshaj

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SUMMARY

Backend Software Engineer with a Master's in Computer Science, experienced in building RESTful APIs, full-stack web applications, and AI-integrated systems. Proficient in Python (Django, Flask, FastAPI), Java (Spring Boot), TypeScript, and PostgreSQL. Thesis research focused on applying Large Language Models to Natural Language Processing and Text-to-SQL translation.

EDUCATION

University of Verona

Master in Computer Science and Engineering (105/110)

Verona, Italy

Oct. 2022 – Oct. 2025

- **Thesis:** *Using LLMs to Translate Natural Language into SQL Queries* — evaluated and compared multiple LLMs (SQLCoder, GPTs, Gemini, Claude) on their ability to generate accurate SQL from natural language, analysing trade-offs in accuracy, latency, and cost across model families.

University of Verona

Bachelor in Computer Science

Verona, Italy

Oct. 2019 – Oct. 2022

- **Thesis:** *Health monitoring and activity suggestion system using Fitbit data and the Apriori algorithm, built with the Ionic Vue framework.*

EXPERIENCE

IT Support Specialist

Generali Italia: Banca e Assicurazioni (Full-time, alongside M.Sc. studies)

Sept. 2024 – Oct. 2025

Verona, Italy

- Provided technical assistance to employees by troubleshooting hardware, software, and network issues, ensuring minimal downtime and smooth daily operations.

Software Developer – Internship

BSQ Security

Jan. 2024 – May 2024

Verona, Italy

- Built a RESTful API using Flask for an AI-powered CCTV alarm detection system, enabling seamless integration with front-end clients and external services.
- Developed a Telegram bot to deliver automated alarm notifications to subscribed users, integrated with the backend API.
- Debugged and performed unit testing of the web application qboard.app using pytest, ensuring functional reliability before release.
- Authored technical documentation to ensure reliability and maintainability of developed components.

Vue Developer – Internship

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Dec. 2022 – Apr. 2023

Verona, Italy

- Developed a web application using Vue.js, TypeScript, Tailwind CSS, and Docker.

TECHNICAL SKILLS

Programming Languages: Java, Python, C, SQL (PostgreSQL), JavaScript, TypeScript

Frameworks: Spring Boot, Django, Flask, FastAPI, Node.js, JUnit, Angular, Vue.js, TailwindCSS

Developer Tools: Git, Docker, VS Code, IntelliJ, Linux, Postman

Libraries: pandas, NumPy, Matplotlib, Transformers, PyTorch, Scikit-learn

Languages: Albanian (Native), English (Fluent) C2, Italian (Fluent) C2

PERSONAL PROJECTS

Albanian Burocracy RAG

Python, ChromaDB, LiteLLM, OpenAI API, Pydantic, Gradio, httpx, Playwright, SQLite, Plotly

Mar. 2026 – Present

- Built an API-first pipeline for crawling the public e-Albania catalog, with hierarchical discovery of categories and services via JSON endpoints, Pydantic normalization, idempotent state on SQLite, and Playwright fallback for dynamically rendered pages.
- Built an advanced RAG system with LLM-based intelligent chunking (headline + summary + verbatim text), embeddings with `text-embedding-3-large` on persistent ChromaDB, and dual-query retrieval with LLM reranking to maximize retrieved context precision.
- Designed a double-track automated evaluation suite measuring retrieval quality via MRR and nDCG, and response quality with LLM-as-a-judge (accuracy, completeness, relevance on a 1–5 scale), visualized on a Gradio dashboard with Plotly.

DataGenFlow – AI Data Pipeline Platform

Python, FastAPI, React, TypeScript, SQLite, Pytest, Docker

Feb. 2026

- Developed full-stack features for dynamic management of LLM and embedding models, introducing default model selection, dedicated REST endpoints, transactional persistence, and React interfaces with real-time state updates.
- Implemented resilient backend logic to ensure data consistency and automatic default model recovery via SQL migrations, transactional upserts, and application-level validations.
- Built unit, integration, and e2e tests with Pytest to verify critical configuration scenarios, automatic reassignment, and complex AI workflows.
- Extended the platform with a conversational data augmentation pipeline, introducing custom AI blocks for semantic infilling, duplicate removal, and structure sampling, plus JSON/template-driven support for dynamic configurations.

Store Inventory Management System

Dec. 2025 – Feb. 2026

Python, Django, Angular, PostgreSQL, Docker

- Developed a full-stack web application with a Django backend, PostgreSQL database, and Angular frontend to manage inventory, sales, and restocking for a retail fish store.
- Developed auto-generation of weekly and monthly reports based on sales data, inventory, customer history, and restocking records.
- Deployed and hosted the application on Hetzner using Docker containers; currently in active use by the store owner.

Hospitium

Jul. 2024

Java, Spring Boot, Gradle, H2

- Developed a software system to manage the operations of a medical clinic, supporting doctors and nurses in handling medications and blood test workflows.
- Implemented a shared secretarial interface for managing appointments and patient records.

Vue Job Search

Sept. 2022

TypeScript, Vue.js, Django, PostgreSQL, REST API

- Developed a job search platform allowing users to filter listings by location, job type, and keywords.
- Implemented user authentication and profile management features.

More projects: github.com/LedjoLleshaj

AWARDS & TRAINING

CyberChallenge.IT

Feb. 2022 – Jun. 2022

cyberchallenge.it

University of Verona, Italy

- Selected through a national competitive test to attend a 4-month cybersecurity training program covering Network Security, Web Application Security, Cryptography, and Ethical Hacking.